

Entry, configuration and run provenance

A single entry layer validates local and HPC configuration before any literature, deployment or evaluation Agent is allowed to execute.

1	4	1	1
unified menu	control files	environment file	run manifest

Agent contracts and responsibilities

versioned Markdown or control files

ROLE	DEFINITION FILE	RESPONSIBILITY
Workflow control	No role-specific Agent	Human workflow selection and authorization
Global safeguards	workflow_guards.py	Preflight checks before deployment or evaluation

Execution sequence

ordered file handoffs

01	02	03	04	05
Select	→ Dispatch	→ Configure	→ Gate	→ Record
run_menu.bat	amp_benchmark_menu.py	config.py + .env	workflow_guards.py	run_manifest.py

Persisted evidence and stage handoff

project-state artifacts

ARTIFACT	ROLE	AUDIT / HANDOFF MEANING
run_menu.bat	entry	starts the interactive workflow
amp_benchmark_menu.py	orchestrator	routes literature, onboarding and benchmark tasks
config.py + .env	configuration	API, HPC, SLURM, model and metric settings
workflow_guards.py	gate	blocks incomplete or unsafe runs
run_manifest.py	provenance	records inputs, versions, events and artifacts

Control invariant

Credentials remain local; every formal run receives a manifest before downstream scientific outputs are accepted.

Evidence search, recommendation and persistent memory

Role-specific Agents retrieve, compress, challenge and reconcile literature evidence before freezing model, dataset and metric recommendation tables.

2,365	304	241	20
papers	evidence batches	compact chunks	recommended models

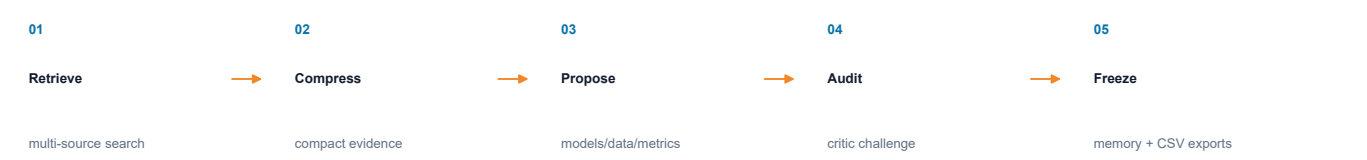
Agent contracts and responsibilities

versioned Markdown or control files

ROLE	DEFINITION FILE	RESPONSIBILITY
Compress	evidence_compressor_agent.md	Retain traceable evidence while reducing context
Extract	info_extractor_agent.md	Produce structured paper and repository metadata
Scout	model_dataset_agent.md	Propose models and benchmark datasets
Metrics	metric_agent.md	Define endpoints and evaluation evidence
Critic	critic_agent.md	Challenge unsupported or incomplete claims
Chief	chief_agent.md	Reconcile decisions and write persistent memory

Execution sequence

ordered file handoffs



Persisted evidence and stage handoff

project-state artifacts

ARTIFACT	ROLE	AUDIT / HANDOFF MEANING
evidence_pool.json	raw evidence	2,365 papers / 304 batches
compact_evidence_pool.json	compressed evidence	2,361 papers / 241 chunks
literature_meeting_screening_decisions.csv	audit decisions	487 identities / 59 accepted
literature_deep_research_memory.json/.md	long-term memory	machine-readable state plus human-readable record
recommended_{models,datasets,metrics}.csv	frozen handoff	20 models / 3 datasets / 44 metric rows

Memory rule

The next literature run reads and updates the stored memory; the recommendation list is versioned evidence, not a permanently fixed template.

Repository registration and HPC deployment

README-derived model contracts are converted into registry records, uploaded to HPC, installed in isolated environments and admitted only after smoke testing.

8	8	42	1
registry entries	marked ready	repair artifacts	smoke-test gate

Agent contracts and responsibilities

versioned Markdown or control files

ROLE	DEFINITION FILE	RESPONSIBILITY
Repository inspector	model_onboarding/repository_inspector_system.md	Translate README, requirements and commands into a registry proposal
Inspector task	model_onboarding/repository_inspector_task.md	Constrain the repository-specific extraction task
Self-heal system	model_execution/self_heal_system.md	Diagnose logs and propose bounded repairs
Self-heal task	model_execution/self_heal_task.md	Retry only approved environment or invocation changes

Execution sequence

ordered file handoffs



Persisted evidence and stage handoff

project-state artifacts

ARTIFACT	ROLE	AUDIT / HANDOFF MEANING
new_model_onboarding.py	local engine	repository inspection and registry generation
hpc_model_ops.py	HPC engine	upload, environment creation, repair and smoke test
data/local_registry.json	durable model state	repository, environment, command and readiness
data/hpc_self_heal/	repair evidence	failure logs, proposed repair and retry record
skip_env_setup=true	admission state	written only after the deployment gate succeeds

Admission rule

A model name in the registry is insufficient; environment readiness, inference command and smoke-test evidence must all be available before formal benchmarking.

Automatic and manual predictions converge at one evaluator

HPC-generated predictions and externally generated prediction tables remain separate until both satisfy the same sample, label and probability schema.

59,311	1,203	1,796	18
C_AMPs-predict test	Veltri test	ProteoGPT test	usable models

Agent contracts and responsibilities

versioned Markdown or control files

ROLE	DEFINITION FILE	RESPONSIBILITY
Architect	runtime_prompts/benchmark_architect.md	Infer model invocation and output schema
Coder	runtime_prompts/coder.md	Generate executable evaluation code
ETL	runtime_prompts/dataset_etl_agent.md	Align sequence IDs, labels and probabilities
Critic	runtime_prompts/critic.md	Audit code, outputs and missingness
PI	runtime_prompts/pi.md	Approve, reject or request a bounded revision

Execution sequence

ordered file handoffs



Persisted evidence and stage handoff

project-state artifacts

ARTIFACT	ROLE	AUDIT / HANDOFF MEANING
data/runs/{run_id}/manifest.json	automatic provenance	HPC inputs, events and retrieved artifacts
data/manual_predictions/{dataset}/	manual provenance	archived source prediction tables
final_results_with_predictions.csv	standard table	aligned sample IDs, labels and probabilities
eval_result.json + scientific_evaluation.json/.md	metric evidence	common metric protocol and statistical summaries
critic_individual.md	independent audit	dataset-specific interpretation and failure review

Evaluation invariant

Automatic and manual branches cannot change the metric definition; both enter the same evaluator only after schema alignment and provenance recording.

Fifty bounded rounds convert metric evidence into consensus ranking

Three specialist proposals are reviewed independently and reconciled by a Chief Agent before each accepted metric vector scores all models across datasets.

12	50	600	900
eligible metrics	decision rounds	weight rows	model scores

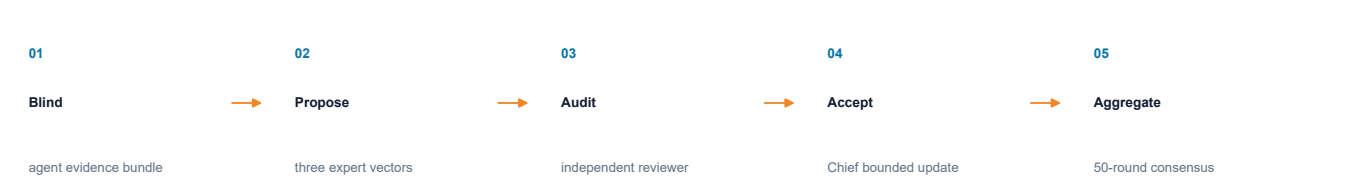
Agent contracts and responsibilities

versioned Markdown or control files

ROLE	DEFINITION FILE	RESPONSIBILITY
Literature	weight_meeting/literature_agent.md	Judge biological relevance and evidence support
Statistics	weight_meeting/statistics_agent.md	Judge validity, uncertainty and metric behavior
Screening	weight_meeting/screening_agent.md	Judge operational robustness and screening utility
Reviewer	weight_meeting/reviewer_agent.md	Audit proposals against blinded evidence
Chief	weight_meeting/chief_agent.md	Accept a bounded weight vector or request revision
Advisor	weight_meeting/research_advisor.md	Interpret stability, complementarity and Top-3

Execution sequence

ordered file handoffs



Persisted evidence and stage handoff

project-state artifacts

ARTIFACT	ROLE	AUDIT / HANDOFF MEANING
agent_evidence_bundle.json	blinded evidence	3 datasets / 18 models / eligible metrics
*_agent_proposals.json	specialist proposals	Literature, Statistics and Screening vectors
reviewer_agent_audit.json	independent audit	proposal-level challenges and decisions
chief_initial_decision.json + rounds/round_001-050.json	decision state	initial vector plus every accepted update
weights/scores/ranking_50_rounds.csv	terminal evidence	current Top-3: pepnet_standard / amplify_imb / C_AMPs-predict

Aggregation equations

Single round: $S_m^{(t)} = \sum_k w_k^{(t)} q_{mk}$; terminal evidence: $\hat{S}_m = 50^{-1} \sum_{t=1}^{50} S_m^{(t)}$. No model-specific bonus is applied.

Exhaustive three-model analysis tests performance complementarity

Every three-model combination is evaluated by equal-probability soft voting and ranked with the stored cross-dataset evidence without model-specific bonuses.

18 eligible models	816 three-model combinations	3 benchmark datasets	1 recommended trio
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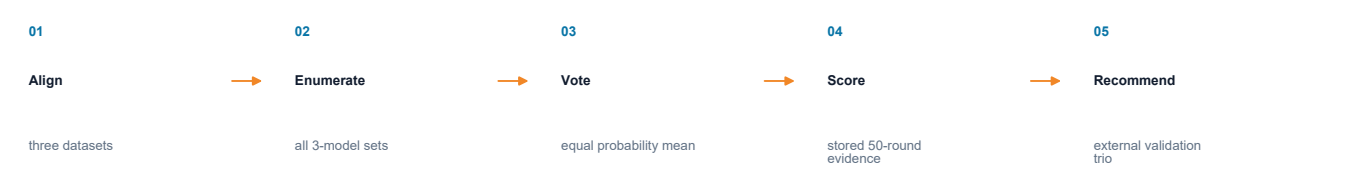
Agent contracts and responsibilities

versioned Markdown or control files

ROLE	DEFINITION FILE	RESPONSIBILITY
Chief handoff	chief_agent.md	Provides the accepted ranking evidence
Research advisor	research_advisor.md	Interprets stability and complementary errors
Combination engine	ensemble_top3_selector.py	Scores every unique three-model combination

Execution sequence

ordered file handoffs



Persisted evidence and stage handoff

project-state artifacts

ARTIFACT	ROLE	AUDIT / HANDOFF MEANING
ensemble_top3_selector.py	selection engine	exhaustive equal-probability soft voting
ensemble_top3_combination_ranking.csv	full ranking	816 evaluated combinations
ensemble_top3_selection.json	selection record	method, datasets, models, ranking and caveat
recommended_models	current trio	C_AMPs-predict / esm-AxP-GDL / pepnet_standard
amp_research_advisor.py	interpretation	performance, stability and future validation plan

Scientific boundary

The recommended trio is exploratory because selection used the same benchmark datasets; a formal performance claim requires an untouched external validation set.

Figures expose evidence without becoming evidence

Publication assets are generated from stored CSV and JSON files, while tests and QA records verify rendering, evaluation and workflow contracts.

4	300	600	24
export formats	PNG dpi	TIFF dpi	test modules

Agent contracts and responsibilities

versioned Markdown or control files

ROLE	DEFINITION FILE	RESPONSIBILITY
Presentation layer	figures/	Generate publication images from stored results
Quality layer	tests/	Check memory, gates, evaluation, manifests and ranking
Protocol	SCIENTIFIC_EVALUATION_PROTOCOL.md	Document metric and interpretation rules

Execution sequence

ordered file handoffs

01	02	03	04	05
Read	→ Render	→ Export	→ Inspect	→ Archive
source CSV/JSON	Python/matplotlib	SVG/PDF/PNG/TIFF	resolution and overlap	source data + QA notes

Persisted evidence and stage handoff

project-state artifacts

ARTIFACT	ROLE	AUDIT / HANDOFF MEANING
figures/*.svg	editable vector	selectable text and editable paths
figures/*.pdf	publication vector	submission and review layout
figures/*.png + *.tiff	raster export	300-dpi preview and 600-dpi submission
source_data_*.csv	traceability	values displayed in each quantitative panel
qa_notes.json + tests/	quality evidence	dimensions, exclusions, caveats and contract tests

Reporting invariant

Plots, captions and screenshots are terminal presentation artifacts; formal numbers must always be traced back to their CSV, JSON, manifest and round-specific decision records.